Sidratul Muntaher Meheraj

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Kabi Sufia Kamal Hall, University of Dhaka

Education

University of Dhaka 2020 – March 2025
B.Sc. in Computer Science and Engineering ***[CGPA: 3.10/4.00]*** *Dhaka, Bangladesh*

- **Relevant Coursework:** Data Structures and Algorithms (C++), Probability & Statistics, Object Oriented Programming (Java), Software Design Patterns (Java), Artificial Intelligence (Python)

Chattogram College 2017 – 2019
Higher Secondary Certificate *Chattogram, Bangladesh*

Research Interests

Machine Learning, Large Language Models (LLMs), Human-Computer Interaction (HCI), AI-assisted Systems

Publications

[**Conference**] Faisal Haque Bappy, Tahrim Hossain, **Sidratul Muntaher Meheraj**, Annoor Sharara Akhand, Tasfia Tabassum, Tarannum Shaila Zaman, Raiful Hasan, Tariqul Islam - *From Preventive to Reactive: How AI Coding Assistants Transform Developers' Security Awareness.* 20th USENIX Symposium on Usable Privacy and Security (SOUPS 2026).

Experience

Dhaka University Research Society Nov 2025 – May 2026
Research Manager *Hybrid*

- Conducted structured interviews with participants and collaborated across teams to prepare comprehensive reports.
- Coordinated research workflows and maintained clear communication between team members and supervisors.

Edulytics Dec 2025 – Feb 2026
Junior Project Coordinator *Hybrid*

- Served as the primary communication bridge between the development team and external stakeholders, ensuring alignment on deliverables and timelines.
- Monitored and reported on project KPIs, ensuring all targets were consistently met and documented.
- Facilitated meetings, tracked action items, and maintained product quality through structured coordination.

Japan-Based Matrimonial Application June 2025 – Oct 2025
Flutter Developer *Remote*

- Contributed to ideation, development, and management of a cross-platform mobile application targeting Japanese and South Asian users.
- Designed and developed a responsive, user-friendly interface to enhance accessibility and user experience across diverse audiences.
- Conducted end-to-end QA testing across user flows and usability scenarios to enhance reliability.

Projects

Co-LAB | *React.js, JavaScript, Python, FastAPI*

- Developed a full-stack web application enabling users to showcase projects, publish blogs, and engage in community discussions using RESTful APIs, JWT authentication, and Redux state management.

Personal Audio Transcriber | *Python, Streamlit*

- Built a transcription tool that extracts audio from YouTube videos and transcribes them using a custom backend model via yt-dlp; deployed on Streamlit for easy URL submission and transcript viewing.

Research Experience

Optimizing University Recommendation System | *Python, Scikit-learn, PyTorch*

- Developed a knowledge-graph-based recommender system using Graph Attention Networks (GAT) and Graph Contrastive Learning (KGCL); addressed cold-start and data sparsity challenges.
- Implemented attention mechanisms to assign higher weights to relevant nodes, and contrastive learning to generate positive/negative pairs for improved model robustness.
- Collected data via web scraping from university-related websites and Facebook groups; generated synthetic datasets where real data was insufficient.

Technical Skills

Languages: Python, C++, Java, SQL, JavaScript
ML / AI: PyTorch, Scikit-learn, PyTorch Geometric, Large Language Models (LLMs), Generative AI, RAG
Backend & Web: FastAPI, Spring Boot, Next.js, React.js, REST APIs
Tools & Database: PostgreSQL, MongoDB, Git, GitHub, Docker, Google Colab